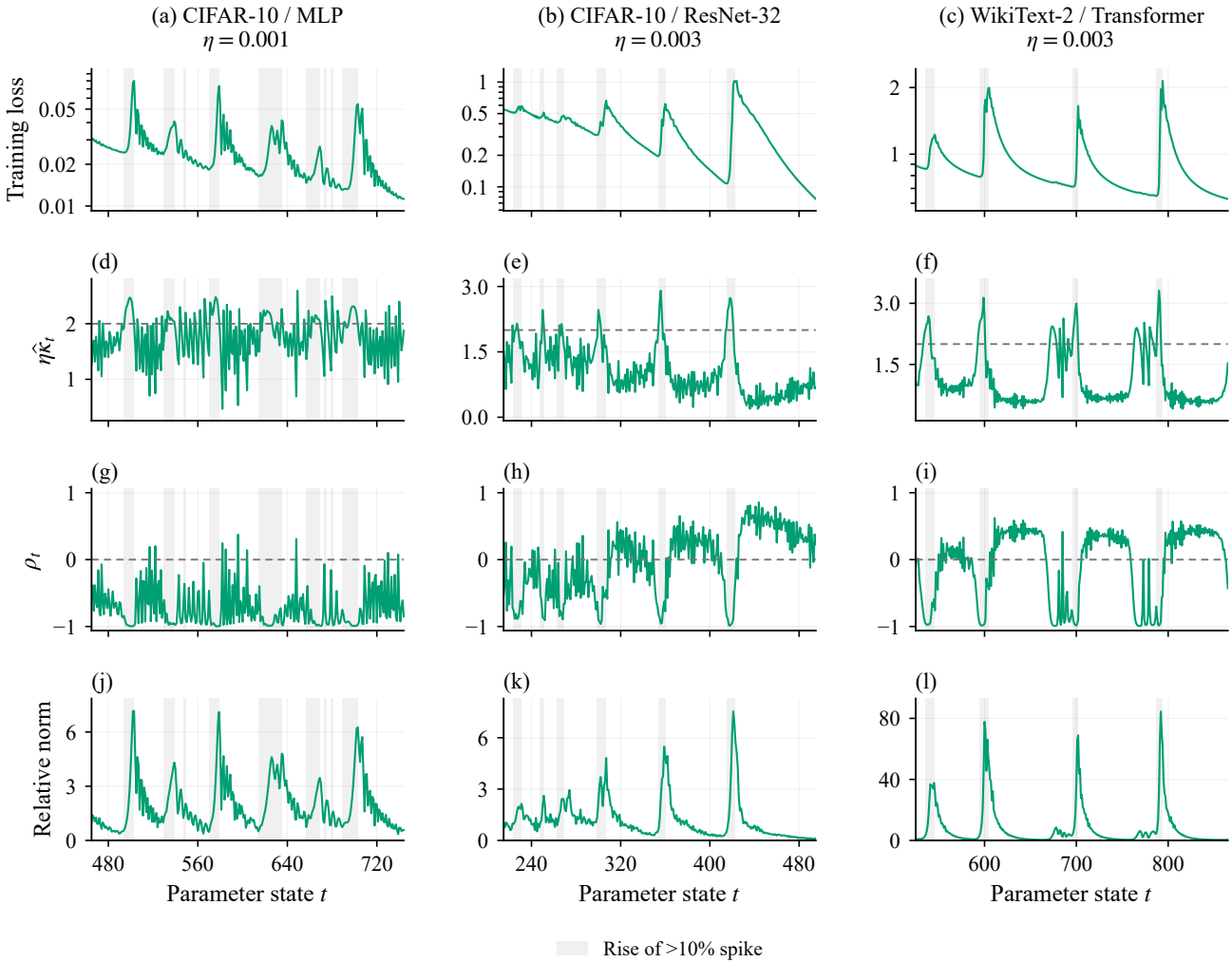

(a) CIFAR-10 / MLP
$\eta = 0.001$
(b) CIFAR-10 / ResNet-32
$\eta = 0.003$
(c) WikiText-2 / Transformer
$\eta = 0.003$
Training loss
0.05
0.02
0.01
1
0.5
0.2
0.1
2
1
(d)
(e)
(f)
$\hat{\eta}\hat{\kappa}_t$
2
1
3.0
1.5
0.0
3.0
1.5
(g)
(h)
(i)
ρ_t
1
0
−1
1
0
−1
1
0
−1
(j)
(k)
(l)
Relative norm
6
3
0
6
3
0
80
40
0
480
560
640
720
240
320
400
480
600
700
800
Parameter state t
Parameter state t
Parameter state t
Rise of >10% spike